

Shadow Play using Ultrasound Levitated Props

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ABSTRACT

Shadow play is a traditional culture to communicate narratives. However, its inheritance is in danger, and traditional methods have limitations in their expression. We propose a novel system to perform a shadow play by levitating props using an ultrasound speaker array. Our system computes an ideal position of levitating props to create a shadow of the desired image. Shadow play will be performed by displaying a sequence of images as shadows. Since the performance is automated, our work makes shadow play accessible to people for generations. Also, our system allows 6 DoF and floating movement of props, which expands the limit of expression. Through this system, we aim to enhance shadow plays informatively and aesthetically.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Displays and imagers.

KEYWORDS

Interactive Systems, Ultrasound Levitation, Shadow Arts

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1 INTRODUCTION

Shadow is the most primitive way to display objects using lights. In an Asian-centered culture called "shadow play," the shadows of articulated cut-out figures called "shadow puppets" are used to tell stories. However, the population of those who could demonstrate shadow play has been getting smaller in recent years. UNESCO added Shadow play to the Intangible Cultural Heritage list in 2011 [2]. Since those with the skills to manipulate shadow puppets are aging, there is a growing demand for operators.

The digitalization of shadow play not only supports its inheritance but also expands the range of narratives shadow play can express. There are various conventional shadow play methods such as human-controlled puppets [1], hand shadowgraphy [3], and robot-controlled puppets [7]. However, in any conventional method, there

is a limit on the range of motion of props. The puppet's mechanical range of motion is the limit of the expression for puppet plays, and the finger's range of motion is the limit for hand shadowgraphy. This limits the range of expression and narratives.

We propose a shadow play system implemented by Ultrasound Levitation Kit, allowing the user to portray dynamic movements and interactions without professional skills. Our shadow play system uses ultrasound levitation to levitate multiple objects. By calculating and manipulating the position of levitated props, the system displays a shadow of the shape of inputted images/videos. Since there is nearly no limitation of the prop's range of motion in using this system, the potential of telling the narrative using shadow plays can be expanded.

Our goals are as follows. 1. We propose a novel way to create a shadow play using levitating objects and ultrasound arrays. 2. We aim to explore the potential of informing the narrative using shadow plays.

To our knowledge, if successfully implemented, it will be the first type of work that creates shadow plays using levitated props. This line of work potentially opens a new avenue of shadow plays in both informative and aesthetic ways.

2 RELATION TO PREVIOUS WORKS

A number of works aim to explore the possibilities of shadow play, especially in the field of Human-Computer Interaction. The common way to create shadow computationally is using robotics technology, such as steering gears [8] and robotic arms [7]. Using robotics allows precise shadow depiction, although it limits the variety of shadows it can portray. For example, the work done by Tang et al. [8] allows depicting humanoid shape shadows but not for other objects such as animals. Also, since the movement is controlled by rigid materials, dynamic shadow animation generation cannot be created. In our shadow play system, the movement is manipulated by levitation; thus, a robust shadow shape generation is possible.

Another common way to manipulate shadows digitally is to create a Virtual Shadow: a fake shadow made by casting lights except for the place where a shadow is pictured. The benefit of this method is that since it can create dynamic shadow movements easily, it can create a shadow that does not exist. Using these characteristics, this method is used to extend the narrative of shadow art [6] and is also used in artists' works [4]. There is a lot to explore in this methodology. However, we still believe that real shadow art possesses value. This is because, in our opinion, the experience of shadow art is about simply looking at the shape of shadow but formed by imagining the objects behind the shadow. This means the existence of the object that creates the shadow brings another value to shadow plays. Thus, we believe the system we propose could bring a different value to virtual shadows.

Although it is not about shadows, we find a project done by Morales et al. [5] has several similarities to our project. This work

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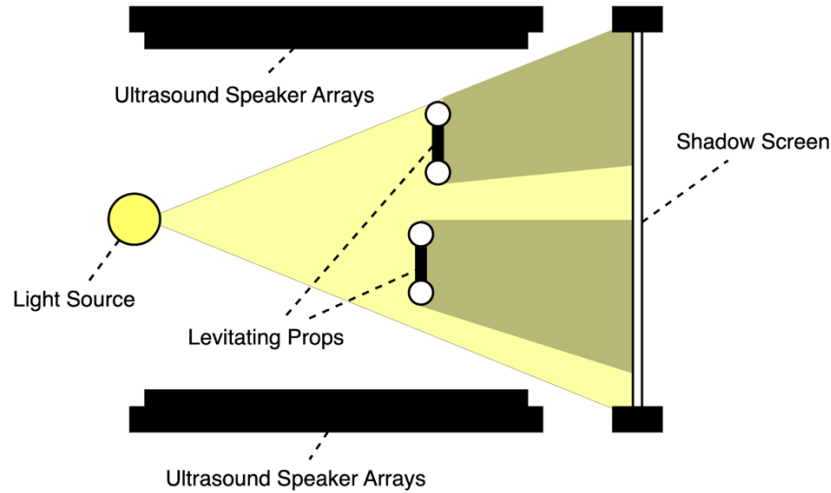


Figure 1: Components of Shadow Play System

proposes a way to manipulate levitated optimized fabric structures using Holographic Acoustic Tweezers. This technology will be the foundation of our project since our project also plans to use fabrics to block the light.

3 SYSTEM COMPONENTS

We aim to implement a system that presents a sequence of target images by shadows of levitated props. As shown in Figure 1, the system will be constructed with a light source, ultrasound speaker arrays, levitating props, and a shadow screen. The system will work on two tasks: designing levitated props and computing placing of levitated props.

3.1 Designing Levitated Props

The first task is to design the props to levitate. Props will be formed of two parts: an intransparent lightweight piece of fabric and multiple beads that act as levitated anchors. When more than 3 anchor beads are attached to the fabric, props will have 6 degrees of freedom, meaning that it could move and rotate in every direction.

According to a study of LeviProps [5], trapping stiffness depends on the position of anchor beads. Their optimization algorithm of anchor position could be applied to our system to design levitated props.

3.2 Computing Placing of Levitated Props

The second task is to compute the position of levitated props that could cast the shadow of the desired image. Using the desired image, position of the screen, light source, and the shapes of props as inputs, we will develop an algorithm to compute the ideal position of levitated props. We will explore the limitations of this device, such as how detailed the shape can be, or how much can this device make the props levitate.

4 APPLICATIONS

4.1 Reconstructing Traditional Shadow Play

The high degree of freedom in our system allows traditional shadow plays to be reconstructed. Image frames of recorded video of shadow play can be used to reconstruct the play using our system.

4.2 Expanding the Limit of Shadow Play

Since our system has higher degrees of freedom than traditional shadow play methods, it could expand the possibilities of performance of shadow play. Figure 2 shows examples of possible applications. For example, shadows of flying characters had to be traditionally created by shadow puppets connected to sticks or strings. In this situation, it was unavoidable to have unnecessary shadows of sticks and strings. However, using our system, shadows of flying characters like birds could be simply created by ultrasound levitated props, leading to a better image without unnecessary shadows. Another example is the dynamic movement of props and shadows, which wasn't traditionally possible due to the limit of the mechanical and physical range of motion. For instance, our system can depict some sort of deformation such as explosion or fluid expression.

4.3 Interactive Shadow Play

Our system could also be applied to create shadow play with rich interaction, which was difficult by traditional methods. For example, if the movement of an audience's body could be tracked by a camera, we could have shadows interact with body movement, or even display a character that syncs with an audience's motion.

5 DEVELOPMENT PLAN

We plan to develop a position estimation algorithm of the props to create a desired shadow shape first. This would be done before

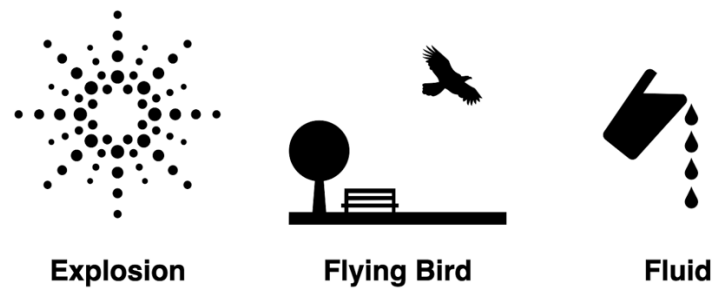


Figure 2: Shadows that are possible with our system

the ultrasound levitation kit arrives. After we get the ultrasound levitation kit, we will design the shape of props, and inspect the limitation of shadow creation. While improving frame rate and quality of shadow play, we will also explore possible applications by creating interactive shadow play performance using our system.

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